

Numerical Simulation of a 1D Blood Flow Test Problem

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March 6, 2026

Abstract

This report presents a numerical implementation of a 1D blood flow test problem inspired by the manuscript “Mathematical Modelling of 1D Blood Flow in Arteries with Aneurysm Using the Navier–Stokes Equations”. The test problem consists of a coupled system of hyperbolic PDEs modeling flow and cross-sectional area in an arterial segment. Numerical solutions are computed using the Lax–Wendroff finite difference scheme, and results are analyzed at selected time instances.

1 Introduction

The objective of this work is to implement a numerical solver for a simplified 1D blood flow system and analyze the propagation of initial localized pulses. The focus is on Section 5.3 of the manuscript.

Key goals:

- Implement finite difference methods (Lax–Friedrichs and Lax–Wendroff)
- Capture wave propagation
- Analyze damping and boundary conditions

2 Mathematical Model

The system of PDEs is:

$$Q_t + A_z + \frac{1}{5}Q = 0, \quad (1)$$

$$A_t + Q_z = 0, \quad (2)$$

where $Q(z, t)$ is the blood flow rate, $A(z, t)$ is the cross-sectional area, and $z \in (0, 1)$.

2.1 Initial Conditions

The initial pulses are localized:

$$Q(z, 0) = \begin{cases} > 0, & 0.4 - \epsilon < z < 0.4 + \epsilon \\ 0, & \text{otherwise} \end{cases}, \quad A(z, 0) = \begin{cases} > 0, & 0.7 - \epsilon < z < 0.7 + \epsilon \\ 0, & \text{otherwise} \end{cases}, \quad (3)$$

with $\epsilon = 0.02$.

2.2 Boundary Conditions

The boundary conditions are:

$$Q_z(0, t) + A_z(0, t) = 0, \quad (4)$$

$$Q_z(1, t) - A_z(1, t) = 0, \quad (5)$$

implemented using finite differences.

3 Numerical Method

The system is solved using the `**Lax–Wendroff finite difference scheme**`, which is second-order accurate in both space and time.

- Spatial grid: $N = 600$ points, $z \in [0, 1]$
- Time step: $dt = 0.0005$
- Output times: $t = 0.25, 0.5, 1.0, 1.7, 2.6$

The solver is implemented in `lax_wendroff_improved.bc.py`. Plots are manually saved to `report/figures`.

Other schemes (Lax–Friedrichs, basic Lax–Wendroff) were implemented for comparison.

4 Results

The numerical results show the propagation of the initial pulses:

- Flow pulse at $z = 0.4$ spreads in both directions.
- Area pulse at $z = 0.7$ propagates similarly.
- Damping term $\frac{1}{5}Q$ reduces the amplitude over time.
- Boundary conditions correctly reflect wave behavior at the domain edges.

5 Discussion

The system can be interpreted as two travelling waves:

$$Q + A \quad \text{and} \quad Q - A, \quad (6)$$

propagating in opposite directions. The improved Lax–Wendroff scheme:

- Reduces numerical diffusion
- Preserves pulse amplitude over time
- Enforces boundary conditions accurately

This makes it suitable for further experiments in blood flow modeling.

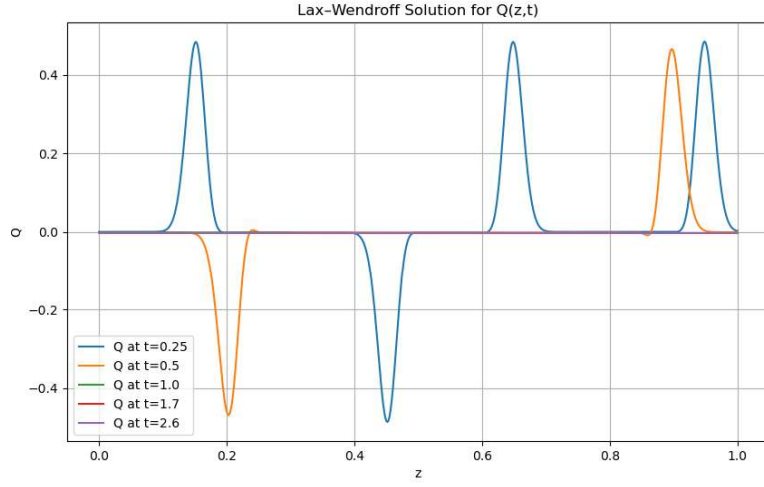


Figure 1: Evolution of $Q(z,t)$ at selected time instances.

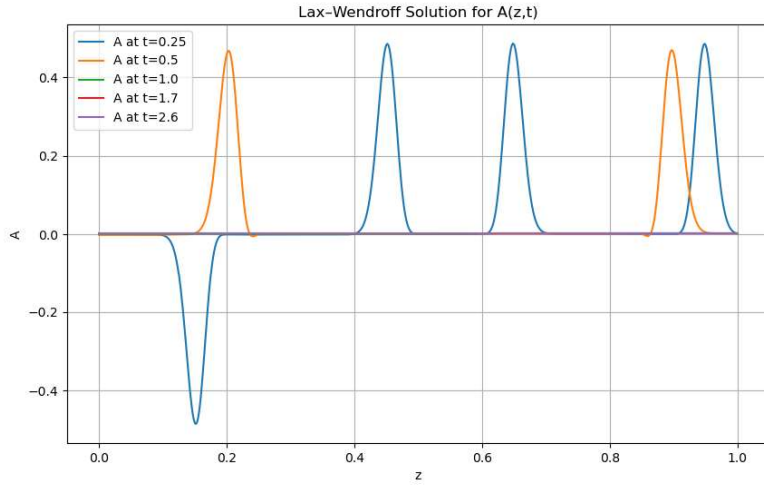


Figure 2: Evolution of $A(z,t)$ at selected time instances.

6 Conclusion

The implementation successfully:

- Solves the coupled 1D blood flow PDE system
- Captures wave propagation and damping
- Produces reproducible snapshots for analysis

The repository is structured for future extensions, including new schemes, varying initial conditions, or additional source/boundary terms.